# CMSC216: x86-64 Control Flow

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# Logistics

## Reading Bryant/O'Hallaron

- Ch 3.6: Control Flow
- Ch 3.7: Procedure calls

## Goals

- RIP: Instruction Pointer
- Jumps and Control flow
- Comparison / Test Instructions
- RIP/Jumps/Compare/Test
- Procedure calls

## Assignments

- Lab07: 2 problems
  1. ASM Stack Manipulation
  2. ASM Global Variables
- HW07: "Stack Smashing"
- P3: Up and due in ~~10~~ 8 days

  1. Clock functions in assembly
  2. Binary debugging

# Announcements

## Midterm Feedback Posted

- Linked at the bottom of the course schedule
- Responses for some free-form feedback is given
- Thanks to all that provided feedback

## Stack Alignment Trickiness

- This week deals with procedure calls and stack alignment a tricky issue for several reasons
- Lab07 involves doing this in `order2` and `order3`
- Stack growth for the `call` instruction will be by 8, 24, 40, 56, ... bytes to align to a 16-byte boundary
- Some processors accept bad alignment, others don't
- Past students have reported "It worked on Zaratan but failed on Gradescope" so be careful

  *Testing can only prove the presence of bugs, not their absence. – Edsger Dijkstra*

# Control Flow in Assembly and the Instruction Pointer

## Instruction Pointer Register

- **`%rip`: special register** (not general purpose) referred to as the **Instruction Pointer** or Program Counter
- `%rip` contains main memory address of next assembly instruction to execute
- After executing an instruction, `%rip` automatically updates to the subsequent instruction

  OR in a Jump instruction, `%rip` changes non-sequentially
- **Do not** add/subtract with `%rip` via `addq`/`subq`: `%rip` automatically updates after each instruction

## Jump Instructions

- **Labels** in assembly indicate jump targets like `.LOOP:`
- **Unconditional Jump**: always jump to a new location by changing `%rip` non-sequentially
- **Comparison / Test**: Instruction, sets `EFLAGS` bits indicating relation between registers/values (greater, less than, equal)
- **Conditional Jump**: Jumps to a new location if certain bits of `EFLAGS` are set by changing `%rip` non-sequentially; otherwise continues sequential execution

# Exercise: Loop Sum with Instruction Pointer (rip)

- Can see direct effects on `rip` in disassembled code
- `rip` increases corresponding to instruction length
- Jumps include address for next `rip`

```c
// C Code equivalent
long sum=0, i=1, lim=100;
while(i<=lim){
  sum += i;
  i++;
}
return sum;
```

```
00000000000005fa <main>:
ADDR  HEX-OPCODES            ASSEMBLY              EFFECT ON RIP
 5fa: 48 c7 c0 00 00 00 00   mov    $0x0,%rax    # rip = 5fa -> 601
 601: 48 c7 c1 01 00 00 00   mov    $0x1,%rcx    # rip = 601 -> 608
 608: 48 c7 c2 64 00 00 00   mov    $0x64,%rdx   # rip = 608 -> 60f
000000000000060f <LOOP>:
 60f: 48 39 d1               cmp    %rdx,%rcx    # rip = 60f -> 612
 612: 7f 08                  jg     61c <END>    # rip = 612 -> 614 OR 61c
 614: 48 01 c8               add    %rcx,%rax    # rip = 614 -> 617
 617: 48 ff c1               inc    %rcx         # rip = 617 -> 61a
 61a: eb f3                  jmp    60f <LOOP>   # rip = 61a -> 60f
000000000000061c <END>:
 61c: c3                     retq   # rip 61c -> return address
```

# Disassembling Binaries

- Binaries hard to read on their own
- Many tools exist to work with them, notably `objdump` on Unix
- Can **disassemble** binary: show "readable" version of contents

```
>> gcc -Og loop.s                  # COMPILE AND ASSEMBLE

>> file a.out
a.out: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV),

>> objdump -d a.out                # DISASSEMBLE BINARY
a.out:     file format elf64-x86-64
...
Disassembly of section .text:
...
0000000000001119 <main>:
    1119:       48 c7 c0 00 00 00 00    mov    $0x0,%rax
    1120:       48 c7 c1 01 00 00 00    mov    $0x1,%rcx
    1127:       48 c7 c2 64 00 00 00    mov    $0x64,%rdx
000000000000112e <LOOP>:
    112e:       48 39 d1                cmp    %rdx,%rcx
    1131:       7f 08                   jg     113b <END>
    1133:       48 01 c8                add    %rcx,%rax
    1136:       48 ff c1                inc    %rcx
    1139:       eb f3                   jmp    112e <LOOP>
000000000000113b <END>:
    113b:       c3                      retq
```

# FLAGS: Condition Codes Register

- Most CPUs have a special register with "flags" for various conditions: each bit is True/False for a specific condition
- In x86-64 this register goes by the following names

| Name | Width | Notes |
|---|---|---|
| `FLAGS` | 16-bit | Most important bits in first 16 |
| `EFLAGS` | 32-bit | Name shown in GDB |
| `RFLAGS` | 64-bit | Not used normally |

- Bits in `FLAGS` register are **automatically** set based on results of other operations like addition / subtraction / etc.

| Bit | Abbrev | Name | Description |
|---|---|---|---|
| 0 | **CF** | Carry flag | Set if last op caused Unsigned overflow |
| 6 | **ZF** | Zero flag | Set if last op yielded a 0 result |
| 7 | **SF** | Sign flag | Set if last op yielded a negative |
| 8 | TF | Trap flag | Used by gdb to stop after one ASM instruction |
| 9 | IF | Interrupt flag | 1: handle hardware interrupts, 0: ignore them |
| 11 | **OF** | Overflow flag | Set if last op caused SIGNED overflow/underflow |

# Comparisons and Tests

Set the EFLAGS register by using comparison instructions

| Name | Instruction | Examples | Notes |
|---|---|---|---|
| Compare | `cmpX B, A` | `cmpl $1,%eax` | Like `if(eax > 1){...}` |
| | Like: `A - B` | `cmpq %rsi,%rdi` | Like `if(rdi > rsi){...}` |
| Test | `testX B, A` | `testq %rcx,%rdx` | Like `if(rdx & rcx){...}` |
| | Like: `A & B` | `testl %rax,%rax` | Like `if(rax){...}` |

- Immediates like $2 must be the first argument B
- B,A are NOT altered with cmp/test instructions
- EFLAGS register IS changed by cmp/test to indicate less than, greater than, 0, etc.

```
### EXAMPLES:
movl $5, %eax          # 5 = 0b0101
cmpl $1, %eax          # [     ] 5-1=4  : No flags
cmpl $5, %eax          # [ZF   ] 5-5=0  : Zero flag
cmpl $8, %eax          # [   SF] 5-8=-3 : Sign flag

testl $0b0110, %eax # [     ] 0101 & 0110 = 0100
testl $0b1010, %eax # [ZF   ] 0101 & 1010 = 0000
```

# Jump Instruction Summary

All control structures implemented using combination of Compare/Test + Jump instructions.

| *Instruction* | *Jump Condition* | *FLAGS* |
|---|---|---|
| `jmp LAB` | Unconditional jump | - |
| `je LAB` | Equal / zero | `ZF` |
| `jz LAB` | | `ZF` |
| `jne LAB` | Not equal / non-zero | `!ZF` |
| `jnz LAB` | | `!ZF` |
| `js LAB` | Negative ("signed") | `SF` |
| `jns LAB` | Nonnegative | `!SF` |
| `jg LAB` | Greater-than signed | `!(SF xor OF) and !ZF` |
| `jge LAB` | Greater-than-equal signed | `!(SF xor OF)` |
| `jl LAB` | Less-than signed | `(SF xor !OF)` |
| `jle LAB` | Less-than-equal signed | `(SF xor !OF) or !ZF` |
| `ja LAB` | Above unsigned | `!CF and !ZF` |
| `jae LAB` | Above-equal unsigned | `!CF` |
| `jb LAB` | Below unsigned | `CF and !ZF` |
| `jbe LAB` | Below-equal unsigned | `CF` |
| `jmp *OPER` | Unconditional jump to variable address | - |

# Examine: Compiler Comparison Inversion

- Often compiler inverts comparisons
- `i < n` becomes `cmpX` / `jge` (jump greater/equal)
- `i == 0` becomes `cmpX` / `jne` (jump not equal)
- This allows "true" case to fall through immediately
- Depending on structure, may have additional jumps
  - `if(){ .. }` usually has a single jump
  - `if(){} else {}` may have a couple

```
## Assembly translation of
## if(rbx >= 2){
##   rdx = 10;
## }
## else{
##   rdx = 5;
## }
## return rdx;
  cmpq  $2,%rbx     # compare: rbx-2
  jl    .LESSTHAN   # goto less than
  ## if(rbx >= 2){
  movq  $10,%rdx    # greater/equal
  ## }
  jmp   .AFTER
.LESSTHAN:
  ## else{
  movq  $5,%rdx     # less than
  ## }
.AFTER:
  ## rdx is 10 if rbx >= 2
  ## rdx is 5 otherwise
  movq  %rdx,%rax
  ret
```

# Logical And / Or in Assembly

Logical boolean operators like `a && b` and `x || y` translate sequences of compare/test instructions followed by conditional jumps. See `andcond_asm.s` and `nestedcond_asm.s`

```c
// andcond.c
int andcond(int edi){
  int ecx;
  if(edi >= 2 && edi <= 10){
    ecx = 10;
  }
  else{
    ecx = 5;
  }
  return ecx;
}
```

C Boolean expressions may "short circuit": never execute code associated with later parts of the condition if early part resolves conditional

```
### andcond_asm.s
.text
.global andcond
andcond:
  cmpl  $2,%edi     # compare: edi-2
  jl .ELSE          #
  cmpl $10, %edi    # compare: edi-10
  jg .ELSE          #

  ## if(edi >= 2 && edi <= 10){
  movl  $10,%ecx    # greater/equal
  ## }
  jmp    .AFTER
.ELSE:
  ## else{
  movl  $5,%ecx     # less than
  ## }
.AFTER:
  movl  %ecx,%eax
  ret
```

# Exercise: The `test` Instruction

```
1  ## jmp_tests_asm.s:
2  main:
3          movl    $0,%eax
4          movl    $5,%edi
5          movl    $3,%esi
6          movq    $0,%rdx
7          movl    $-4,%ecx
8
9          testl   %edi,%edi
10         jnz     .NONZERO
11         addl    $20,%eax
12
13 .NONZERO:
14         testl   %esi,%esi
15         jz      .FALSEY
16         addl    $30,%eax
17
18 .FALSEY:
19         testq   %rdx,%rdx
20         je      .ISNULL
21         addl    $40,%eax
22
23 .ISNULL:
24         testl   %ecx,%ecx
25         jns     .NONNEGATIVE
26         addl    $50,%eax
27
28 .NONNEGATIVE:
29         ret
```

- `testl %eax,%eax` uses bitwise AND to examine a register
- Selected by compiler to check for zero, `NULL`, negativity, etc.
- Followed by `je` / `jz` / `jne` / `jnz` / `js` / `jns`
- Trace the execution
- Determine final value in %eax
- Will demo `jmp_tests_asm.s` in GDB to show how `EFLAGS` changes

# **Answers**: The `test` Instruction

```
1  ### jmp_tests_asm_commented.s:
2  main:
3          movl    $0,%eax         # eax is 0
4
5          movl    $5,%edi         # set initial vals
6          movl    $3,%esi         # for registers to
7          movq    $0,%rdx         # use in tests
8          movl    $-4,%ecx
9
10         ## eax=0, edi=5, esi=3, edx=NULL, ecx=-4
11         testl   %edi,%edi       # any bits set?
12         jnz     .NONZERO        # jump on !ZF (zero flag), same as jne
13         ## if(edi == 0){
14         addl    $20,%eax
15         ## }
16 .NONZERO:
17         testl   %esi,%esi       # any bits set?
18         jz      .FALSEY         # jump on ZF same as je
19         ## if(esi){
20         addl    $30,%eax
21         ## }
22 .FALSEY:
23         testq   %rdx,%rdx       # any bits set
24         je      .ISNULL         # same as jz: jump on ZF
25         ## if(rdx != NULL){
26         addl    $40,%eax
27         ## }
28 .ISNULL:
29         testl   %ecx,%ecx       # sign flag set on test to indicate negative results
30         jns      .NONNEGATIVE   # jump on !SF (not signed; e.g. positive)
31         ## if(ecx < 0){
32         addl    $50,%eax
33         ## }
34 .NONNEGATIVE:
35         ## eax is return value
36         ret
```

```
// Rough C translation is
if(edi == 0){
  eax += 20;
}
if(esi){
  eax += 30;
}
if(rdx != NULL){
  eax += 40;
}
if(ecx < 0){
  eax += 50;
}
```

# (Optional) cmov Family: Conditional Moves

- Instruction family which copies data conditioned on FLAGS[1]
- Can limit jumping in simple assignments

```
cmpq    %r8,%r9
cmovge  %r11,%r10  # if(r9 >= r8) { r10 = r11 }
cmovg   %r13,%r12  # if(r9 >  r8) { r12 = r13 }
```

- Note flags set on **all Arithmetic Operations**
- `cmpX` is like `subQ`: both set FLAG bits the same
- Greater than is based on the SIGN flag indicating subtraction would be negative allowing the following:

```
subq    %r8,%r9    # r9 = r9 - r8
cmovge  %r11,%r10  # if(r9 >= 0) { r10 = r11 }
cmovg   %r13,%r12  # if(r9 >  0) { r12 = r13 }
```

[1] Other architectures like ARM have conditional versions of many instructions like `addlt r1, r2, r3` ; RISC V ditches the FLAGS register in favor of jumps based on comparisons like `BLT x0, x1, LOOP`

# Procedure Calls

Have seen basics so far:

```
main:
        ...
        call  my_func  # call a function
        ## arguments in %rdi, %rsi, %rdx, etc.
        ## control jumps to my_func, returns here when done
        ...

my_func:
        ## arguments in %rdi, %rsi, %rdx, etc.
        ...
        movl  $0,%eax   # set up return value
        ret             # return from function
        ## return value in %rax
        ## returns control to wherever it came from
```

Need several additional notions

- Control Transfer to called function?
- Return back to calling function?
- Stack alignment and conventions
- Register conventions

# Procedure Calls Return to Arbitrary Locations

- `call` instructions always transfer control to start of `return_seven` at line 4/5, like `jmp` instruction which modifies `%rip`
- `ret` instruction at line 6 must transfer control to **different locations**
  1. `call`-ed at line 11 `ret` to line 12
  2. `call`-ed at line 17 `ret` to line 18

  `ret` cannot be a normal `jmp`
- To enable return to multiple places, record a **Return Address** when `call`-ing, use it when `ret`-urning

```
1  ### return_seven_asm.s
2  .text
3  .global return_seven
4  return_seven:
5        movl    $7, %eax
6        ret     ## jump to line 12 OR 18??
7  .global main
8  main:
9        subq    $8, %rsp
10
11       call    return_seven  ## to line 5
12       leaq    .FORMAT_1(%rip), %rdi
13       movl    %eax, %esi
14       movl    $0, %eax
15       call    printf@PLT
16
17       call    return_seven  ## to line 5
18       leaq    .FORMAT_2(%rip), %rdi
19       movl    %eax, %esi
20       movl    $0, %eax
21       call    printf@PLT
22
23       addq    $8, %rsp
24       movl    $0, %eax
25       ret
26 .data
27 .FORMAT_1: .asciz "first:  %d\n"
28 .FORMAT_2: .asciz "second: %d\n"
```

# `call` / `ret` with Return Address in Stack

## `call` Instruction

1. Push the "caller" **Return Address** onto the stack
   Return address is for instruction after `call`
2. Change `rip` to first instruction of the "callee" function

## `ret` Instruction

1. Set `rip` to Return Address at top of stack
2. Pop the Return Address off to shrink stack

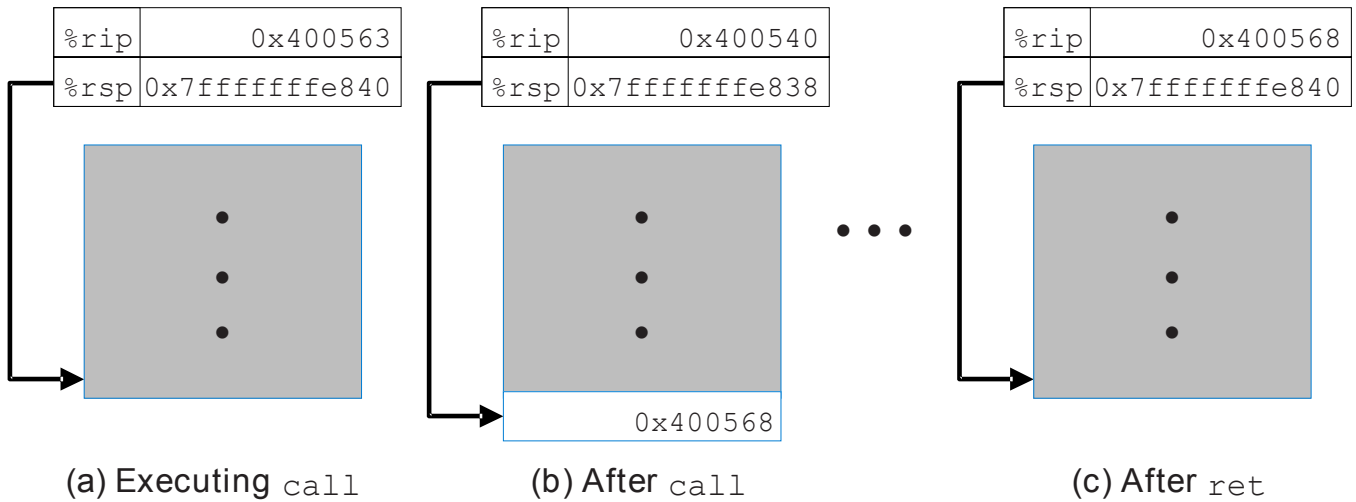



Figure: Bryant/O'Hallaron Fig 3.26 demonstrates call/return in assembly

# return_seven_asm.s 1/2: Control Transfer with call

```
### BEFORE CALL
return_seven:
   0x555555555139 <return_seven>   mov    $0x7,%eax
   0x55555555513e <return_seven+5> retq

main: ...
   0x55555555513f <main>           sub    $0x8,%rsp
=> 0x555555555143 <main+4>         callq  0x555555555139 <return_seven>
   0x555555555148 <main+9>         lea    0x2ee1(%rip),%rdi
   0x55555555514f <main+16>        mov    %eax,%esi


(gdb) stepi
rsp = 0x7fffffffe450 -> call -> 0x7fffffffe448   # push on return address
rip = 0x555555555143 -> call -> 0x555555555139   # jump control to procedure

### AFTER CALL
return_seven:
=> 0x555555555139 <return_seven>   mov    $0x7,%eax
   0x55555555513e <return_seven+5> retq

main: ...
   0x55555555513f <main>           sub    $0x8,%rsp
   0x555555555143 <main+4>         callq  0x555555555139 <return_seven>
   0x555555555148 <main+9>         lea    0x2ee1(%rip),%rdi
   0x55555555514f <main+16>        mov    %eax,%esi

(gdb) x/gx $rsp                     # stack grew 8 bytes with call
0x7fffffffe448: 0x0000555555555148  # return address in main on stack
```

# return_seven_asm.s 2/2: Control Transfer with ret

```
### BEFORE RET
return_seven:
   0x555555555139 <return_seven>   mov    $0x7,%eax
=> 0x55555555513e <return_seven+5> retq

main: ...
   0x55555555513f <main>           sub    $0x8,%rsp
   0x555555555143 <main+4>         callq  0x555555555139 <return_seven>
   0x555555555148 <main+9>         lea    0x2ee1(%rip),%rdi
   0x55555555514f <main+16>        mov    %eax,%esi

(gdb) x/gx $rsp
0x7fffffffe448: 0x0000555555555148  # return address pointed to by %rsp

(gdb) stepi                                          # EXECUTE RET INSTRUCTION
rsp = 0x7fffffffe448 -> ret -> 0x7fffffffe450        # pops return address off
rip = 0x55555555513e -> ret -> 0x555555555148        # sets %rip to return address

### AFTER RET
return_seven:
   0x555555555139 <return_seven>   mov    $0x7,%eax
   0x55555555513e <return_seven+5> retq

main: ...
   0x55555555513f <main>           sub    $0x8,%rsp
   0x555555555143 <main+4>         callq  0x555555555139 <return_seven>
=> 0x555555555148 <main+9>         lea    0x2ee1(%rip),%rdi
   0x55555555514f <main+16>        mov    %eax,%esi

(gdb) print $rsp     -->  $3 = 0x7fffffffe450
```

# **Warning:** `%rsp` is important for returns

- When a function is about to return `%rsp` MUST refer to the memory location of the return address
- `ret` uses value pointed to `%rsp` as the return address
- Segmentation Faults often occur if `%rsp` is NOT the return address: attempt to fetch/execute instructions out of bounds
- Stack is often used to store local variables, stack pointer `%rsp` is manipulated via `pushX` / `subq` instructions to grow the stack.
- Before returning MUST shrink stack and restore `%rsp` to its original value via `popX` / `addq` instructions
- There are computer security issues associated stack-based return value we will discuss later

# Messing up the Return Address

```
### return_seven_buggy_asm.s
.text
.global return_seven
return_seven:
   pushq   $0x42       # push but no pop before returning
   movl    $7, %eax
   ret                 # %rsp points to a 0x42 return address - BAD!
```

```
| REG |   VALUE |     | ADDRESS |    VALUE | NOTE         |
|-----+---------|     |---------+----------+--------------|
| rax |       7 |     | 0x77128 | 0x554210 | Ret Address  |
| rsp | 0x77120 |--->| 0x77120 |     0x42 | Pushed Val   |
```

```
> gcc -g return_seven_buggy_asm.s

> ./a.out
Segmentation fault (core dumped)   ## definitely a memory problem

> valgrind ./a.out                 ## get help from Valgrind
...
==2664132== Jump to the invalid address stated on the next line
==2664132==    at 0x42: ???        ## execute instruction at address 0x42??
==2664132==    by 0x109149: ??? (return_seven_buggy_asm.s:18)
==2664132==  Address 0x42 is not stack'd, malloc'd or (recently) free'd
```

*Valgrind reports like this often indicate failure to restore the stack pointer as happened here. If the stack grows, shrink it before returning.*

# Stack Alignment

- According to the strict x86-64 ABI, must align `rsp` (stack pointer) to 16-byte boundaries when calling functions
- Will often see arbitrary pushes or subtractions to align
  - Functions called with 16-byte alignment
  - `call` pushes 8-byte Return Address on the stack
  - At minimum, must grow stack by 8 bytes to `call` again
- `rsp` changes must be undone prior to return

```
main:                           # enter with at 8-byte boundary
    subq    $8, %rsp            # align stack for func calls
    ...
    call    sum_range           # call function
    ...
    addq    $8, %rsp            # remove rsp change
    ret
```

- Skipping stack alignment works on some machines and breaks on others due to differing processor internals

  *"I passed all the tests on Zaratan but a bunch failed on Gradescope! What gives?"*

  *– Common lament of past students*

# x86-64 Register/Procedure Convention

- Used by `Linux/Mac/BSD/General Unix`
- Params and return in registers if possible

## Parameters and Return

| | |
|---|---|
| RetVal | `rax / eax / ax / al` |
| Arg 1 | `rdi / edi / di / dil` |
| Arg 2 | `rsi / esi / si / sil` |
| Arg 3 | `rdx / edx / dx / dl` |
| Arg 4 | `rcx / ecx / cx / cl` |
| Arg 5 | `r8 / r8d / r8w / r8b` |
| Arg 6 | `r9 / r9d / r9w / r9b` |
| Arg 7 | Push into the stack |
| Arg 8 | Push into the stack |
| ... | ... |

C function prototype indicates number, order, type of args so it is known which registers args will be in

```
int myfunc(char *cp,
           int a, long b);
```

## Caller/Callee Save

**Caller save** registers: alter freely

```
rax rcx rdx rdi rsi
r8  r9  r10 r11      # 9 regs
```

**Callee** save registers: must restore these before returning

```
rbx rbp r12 r13 r14
r15                  # 6 regs
```

**Stack Pointer**: special considerations discussed in detail

```
rsp                  # 1 reg
```

# Caller and Callee Save Register Mechanics

```
main:          # main: the calleR
    ...
    movq $21, %rdi    # calleR save arg 1
    movq $31, %rsi    # calleR save arg 2
    movq $41, %r10    # calleR save
    movq $7,  %rbx    # calleE save
    movq $11, %r12    # calleE save

    call foo  # foo: the calleE

    ## | ?  | %rdi | calleR save arg 1 |
    ## | ?  | %rsi | calleR save arg 2 |
    ## | ?  | %r10 | calleR save       |
    ## | 7  | %rbx | calleE save       |
    ## | 11 | %r12 | calleE save       |

    cmpq $21, %rdi  # unpredictable
    cmpq $7,  %rbx  # predictably equal

    # main MUST restore %rbx and %r12 to
    # original values as function above
    # main() expects them to be unchanged
```

## CalleR Save Regs

May all change across function call boundaries. Not a problem for **Leaf Functions** which do not call any other funcs

## CalleE Save Regs

Have the same values in them after a function call
Using them requires saving their original values in the stack and restoring them

## `sumrange_asm.s`

Full example of callee save regs like `sumrange_c.c`

# Pushing and Popping the Stack

- If local variables or callee save regs are needed on the stack, can use `push` / `pop` for these
- Push and Pop Instructions are compound: manipulate `%rsp` and move data in single instruction

| | |
|---|---|
| `pushX data` | **Grow Stack, store data at top** |
| `pushq %rax` | Like: `subq $8,%rsp; movq %rax,(%rsp)` |
| `pushl $25` | Like: `subq $4,%rsp; movq $25, (%rsp)` |
| | |
| `popX data` | **Shrink Stack, restore data from it** |
| `popl %edi` | Like: `movl (%rsp),%edi; addq $4,%rsp;` |
| `popq %rax` | Like: `movq (%rsp),%rax; addq $8,%rsp;` |

```
main:
    pushq   %rbp            # save register, aligns stack
                            # like subq $8,%rsp; movq %rbp,(%rsp)
    call    sum_range       # call function
    movl    %eax, %ebp      # save answer
    ...
    call    sum_range       # call function, ebp not affected
    ...
    popq    %rbp            # restore rbp, shrinks stack
                            # like movq (%rsp),%rbp; addq $8,%rsp
    ret
```

# Exercise: Local Variables which need an Address

## Compare code in files

- `swap_pointers.c` : familiar C code for swap via pointers
- `swap_pointers_asm.s` : hand-coded assembly version

## Determine the following

1. Where are `main()` local variables `x,y` stored in assembly?
2. Where does the assembly version "grow" the stack?
3. How are the values in `main()` passed as arguments to `swap_ptr()`?
4. Where are `swap_ptr()` local variables stored in assembly?
5. Where does the assembly version "shrink" the stack?

# Exercise: Local Variables which need an Address

1. Where are `main()` local variables `x,y` stored in assembly?
2. Where does the assembly version "grow" the stack?
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4. Where are `swap_ptr()` local variables stored in assembly?
5. Where does the assembly version "shrink" the stack?

```c
// swap_pointers.c
#include <stdio.h>

void swap_ptr(int *a, int *b){
  int tmp = *a;
  *a = *b;
  *b = tmp;
  return;
}

int main(int argc, char *argv[]){
  int x = 19;
  int y = 31;
  swap_ptr(&x, &y);
  printf("%d %d\n",x,y);
  return 0;
}
```

```asm
# swap_pointers_asm.s
.text
.global swap_ptr
swap_ptr:
        movl    (%rdi), %eax
        movl    (%rsi), %edx
        movl    %edx, (%rdi)
        movl    %eax, (%rsi)
        ret
.global main
main:
        subq    $8, %rsp
        movl    $19, (%rsp)
        movl    $31, 4(%rsp)
        movq    %rsp, %rdi
        leaq    4(%rsp), %rsi
        call    swap_ptr

        leaq    .FORMAT(%rip), %rdi
        movl    (%rsp), %esi
        movl    4(%rsp), %edx
        movl    $0, %eax
        call    printf@PLT

        addq    $8, %rsp
        movl    $0, %eax
        ret
.data
.FORMAT:
        .asciz "%d %d\n"
```

# **Answers**: Local Variables which need an Address

1. Where are `main()` local variables `x,y` stored in assembly?
2. Where does the assembly version "grow" the stack?
3. How are the values in `main()` passed as arguments to `swap_ptr()`?

```
// C CODE
int x = 19, y = 31;
swap_ptr(&x, &y)  // need main mem addresses for x,y
### ASSEMBLY CODE
main:                           # main() function
    subq    $8, %rsp            # grow stack by 8 bytes
    movl    $19, (%rsp)         # move 19 to local variable x
    movl    $31, 4(%rsp)        # move 31 to local variable y
    movq    %rsp, %rdi          # address of x into rdi, 1st arg to swap_ptr()
    leaq    4(%rsp), %rsi       # address of y into rsi, 2nd arg to swap_ptr()
    call    swap_ptr            # call swap function
```

4. Where are `swap_ptr()` local variables stored in assembly?

   Stored entirely in registers; args `%rdi,%rsi` are `a,b` while `tmp` is spread across regs `%eax,%edx`

5. Where does the assembly version "shrink" the stack?

```
    addq    $8, %rsp            # shrink stack by 8 bytes
    movl    $0, %eax            # set return value
    ret
```

# Diagram of Stack Variables

- Compiler determines if local variables go on stack
- If so, calculates location as `rsp` + offsets

```c
// C Code: locals.c
int set_buf(char *b, int *s);
int main(){
  // locals re-ordered on
  // stack by compiler
  int size = -1;
  char buf[16];
  ...
  int x = set_buf(buf, &size);
  ...
}
```

| REG | VALUE | Name |
|---|---|---|
| rsp | #1024 | top of stack during main |
| MEM | | |
| ... | ... | ... |
| #1031 | h | buf[3] |
| #1030 | s | buf[2] |
| #1029 | u | buf[1] |
| #1028 | p | buf[0] |
| #1024 | -1 | size |

```asm
## EQUIVALENT ASSEMBLY
main:
    subq    $24, %rsp       # space for buf/size and stack alignment
    movl    $-1,(%rsp)      # retAddr:8, locals: 20, padding: 4, tot: 32
    ....                    # initialize buf and size: main line 6
    leaq    4(%rsp), %rdi   # address of buf  arg1
    leaq    0(%rsp), %rsi   # address of size arg2
    call    set_buf         # call function, aligned to 16-byte boundary
    movl    %eax,%r8        # get return value
    ...
    addq    $24, %rsp       # shrink stack size
```

# Summary of Procedure Calls: ABC() calls XYZ()

| | | | |
|---|---|---|---|
| ABC() | Caller | `callq XYZ` | `# ABC to XYZ` |
| XYZ() | Callee | `retq` | `# XYZ to ABC` |

1. ABC() "saves" any Caller Save registers it needs by either copying them into Callee Save registers or pushing them into the stack
2. ABC() places up to 6 arguments in `%rsi`, `%rdi`, `%rdx`, `...`, remaining arguments in stack
3. ABC() ensures that stack is "aligned": `%rsp` contains an address that is evenly divisible by 16
4. ABC() issues the `callq ABC` instruction which (1) grows the stack by subtracting 8 from `%rsp` and copies a return address to that location and (2) changes `%rip` to the staring address of `func`
5. XYZ() now has control: `%rip` points to first instruction of XYZ()
6. XYZ() may issue `pushX val` instructions or `subq N,%rsp` instructions to grow the stack for local variables
7. XYZ() may freely change Caller Save registers BUT Callee Save registers it changes must be restored prior to returning.
8. XYZ() must shrink the stack to its original position via `popX %reg` or `addq N,%rsp` instructions before returning.
9. XYZ() sets `%rax` / `%eax` / `%ax` to its return value if any.
10. XYZ() finishes, issues the `retq` instruction which (1) sets the `%rip` to the 8-byte return address at the top of the stack (pointed to by `%rsp`) and (2) shrinks the stack by doing `addq $8,%rsp`
11. ABC() function now has control back with `%rip` pointing to instruction after `call XYZ`; may have a return value in `%rax` register
12. ABC() must assume all Caller Save registers have changed

# History: Base Pointer rbp was Previously Special Use

```c
int bar(int, int, int);
int foo(void) {
  int x = bar(1, 2, 3);
  return x+5;
}
```

- 32-bit x86 / IA32 assembly used rbp and rsp to describe stack frames
- All function args pushed onto the stack when calling, changes both Stack/Base Pointer
- x86-64: optimizes %rbp for general purpose use, points at stack only in special situations

```
  # Old x86 / IA32 calling sequence: set both %esp and %ebp for function call
  # Push all arguments into the stack; DON'T DO THIS IN MODERN CODE
  foo:
      pushl %ebp              # modifying ebp, save it
      ## Set up for function call to bar()
      movl  %esp,%ebp         # new frame for next function
      pushl 3                 # push all arguments to
      pushl 2                 # function onto stack
      pushl 1                 # no regs used
      call bar                # call function, return val in %eax
      ## Tear down for function call bar()
      movl %ebp,%esp          # restore stack top: args popped
      ## Continue with function foo()
      addl 5,%eax             # add onto answer
      popl %ebp               # restore previous base pointer
      ret
```